

Investigating the Effect of Temperature on the Apparent Rate Constant of the Decomposition of Aqueous Hydrogen Peroxide.

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IB Chemistry
Internal Assessment

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1 Introduction (493 words)

The decomposition reaction of hydrogen peroxide to form oxygen and water is a thermodynamically favourable process, but unless catalyzed, it proceeds at an extremely slow rate at room temperature.



In this investigation, potassium iodide was used as a catalyst so that the reaction (specifically, the formation of $\text{O}_{2(\text{g})}$) inside the system can occur at a measurable rate within a practical time frame. This investigation aims to explore how temperature impacts the apparent rate constant for the catalyzed decomposition reaction of aqueous hydrogen peroxide. The rate of the reaction is observed indirectly by measuring the rate of increase in gas pressure inside a sealed flask, as $\text{O}_{2(\text{g})}$ builds up.



The collision theory states that a chemical reaction can only occur when particles collide with sufficient energy and the correct orientation. As the temperature of the reaction increase, reacting particles have greater average kinetic energy. This increase in the particle's collision frequency and the increase in the probability of collisions with energy *equal or greater than* the activation energy (E_A). Thus, increasing the temperature of the reaction is expected to increase the rate of the reaction. The temperature dependence of a rate constant can be modelled by the Arrhenius equation:

$$k = A \cdot e^{-\frac{E_A}{RT}} \quad (3)$$

Where k is the rate constant, A is the frequency factor, E_A is the activation energy, R is the gas constant ($8.31 \text{ J K}^{-1} \text{ mol}^{-1}$)¹, and T is the temperature (Kelvins). Taking the natural logarithm on both sides of the equation will yield a linear relationship:

$$\ln(K) = -\frac{E_A}{RT} + \ln(A) \quad (4)$$

Where the slope of the linear graph is $-\frac{E_A}{R}$ and the y-intercept is $\ln(A)$. As this experiment uses pressure measurements rather than concentration measurements, the ideal gas law can be used to relate pressure to the amount of oxygen produced:

$$PV = nRT \quad (5)$$

¹Value obtained from IB Chemistry Data Booklet, version 1.0 (first assessment 2025).

In a sealed flask, the gas volume is considered as constant. The equation is re-ranged to show that moles of gas depends on both pressure and temperature:

$$n = \frac{PV}{RT} \quad (6)$$

Then, for a given ΔP , the change in moles is:

$$\Delta n(\text{O}_2) = \frac{V_{\text{gas}}}{RT} \cdot \Delta P \quad (7)$$

Where T is the target temperature (K) for a specific trial, and V_{gas} is the gas volume inside the sealed flask ($V_{\text{gas}} = V_{\text{total volume}} - V_{\text{solution}}$).

* * *

As a result, the research question for this investigation is:

How does the temperature affect the apparent rate constant for the KI-catalyzed decomposition of aqueous hydrogen peroxide, determined from the rate of increase of gas pressure in a sealed flask?

* * *

Given that the rate constant of the decomposition of aqueous H_2O_2 is expected change with temperature, increasing T should increase the fraction of particles reacting with enough energy to overcome the E_A , so that a greater proportion of particles can collide, leading to a reaction. Though KI acts as a catalyst (provides an alternative pathway that has a lower E_A), the elementary steps still needs to overcome an E_A . Thus, the rate constant k should increase with temperature, according to Arrhenius's equation (approximate linear relationship of $\ln(k)$ vs. $1/T$).

2 Experimental Design (165 words)

This experiment was designed to isolate the temperature of the water bath as the sole variable affecting the pressure for the KI-catalyzed decomposition reaction of aqueous H_2O_2 . Oxygen gas produced during the reaction increased the pressure inside a sealed 250 mL flask. A water bath was used to equilibrate the reaction system to within 1°C of the target temperature. The masses of KI were pre-measured and recorded before data collection, and sealed with plastic wrap and rubber band to prevent exposure to moisture.

Independent variable: Temperature of the reaction system. The H_2O_2 in the sealed flask was equilibrated to the target temperature using a water bath before the start of each trial. The temperature was monitored using a Vernier Temperature Probe.

Dependent variable: Gas pressure in the sealed flask (kPa).

Controlled variable: Initial amount of reactants, total volume of gas space in the sealed flask, environmental conditions, sampling settings, and apparatus setup were all kept constant to reduce variations in the rate.

2.1 Materials (63 words)

Apparatus.

- 1000 mL beaker
- (2x) 600 mL beakers
- 250 mL flask
- 250 mL beaker
- 100 mL beaker
- Glass bowl
- Rubber stoppers
- Vernier equipments (Gas Pressure Sensor, Temperature Probe, LabQuest)
- 25 mL Pipette, suction bulb
- 10 mL graduated cylinder
- Digital balance
- Metal stick from ring stand
- Safety goggles
- Lab coat

Reagents.

- 3% hydrogen peroxide, H_2O_2 (aqueous)
- Potassium iodide, KI (solid)
- Distilled water

2.2 Procedure (614 words)

2.2.1 Setup.

1. Gather the required items from *section 2.1* to work station.
2. Connect the gas pressure sensor to the LabQuest and attach the plastic tubing to the sensor and rubber

stopper.

3. Secure the 250 mL flask on the ring stand using a ring clamp so that it remains stable throughout data collection.
4. Add 25 mL of water to the 250 mL flask (to represent H_2O_2 during setup).
5. Place the glass bowl on the hot plate and add water so that it submerges the 25 mL water line inside the flask.
6. Draw a horizontal line on the neck of the reaction flask to standardize the insertion depth of the rubber stopper in every trial.
 - This ensures that the headspace gas volume (V_{gas}) remain constant between trials.
 - Before inserting the rubber stopper each time, add a few drops of water to the rubber stopper to improve the sealing.
7. Ensure the temperature probe is not touching the bottom of the glass bowl or the sides of the flask. Secure the temperature probe using a ring clamp.
8. Set the LabQuest to record at *5 samples / second*, for a total of *60 seconds* for each trial.

2.2.2 Prepare materials.

1. Pour aqueous H_2O_2 into a clean 600 mL beaker to use as the master solution. Excess can be poured back later using a funnel.
2. Prepare KI for each trial:
 - (a) Weigh 0.30 g of $\text{KI}_{(\text{s})}$ into individual plastic cups, and label each cup with its corresponding temperature level and trial number.
 - (b) Seal each cup using a piece of plastic film, and secure it with rubber band to reduce exposure to moisture from air.
3. Fill another clean 600 mL beaker with ice to lower the temperature of the water bath.

2.2.3 Calibration.

1. Open the valve on the rubber stopper, start recording the atmospheric pressure for 10 seconds, and record the average pressure (kPa).
2. Wash the 25 mL pipette with $\text{H}_2\text{O}_{2(\text{aq})}$ to avoid contamination.

Note: An airtight-sealing calibration was not conducted because the experiment relies on finding ΔP over time. Pressure would be consistently lower than the actual pressure in the case of a minor leak, which does not affect the rate of change in pressure.

2.2.4 Experiment.

1. Adjust the water bath temperature to the target temperature.
2. Pipette 25 mL of H_2O_2 to a clean 250 mL beaker. Once the temperature of the water bath reaches within 1 °C of the target temperature, pour the H_2O_2 into the reaction flask. Keep the reaction flask in the water bath to maintain consistency.
3. Measure 5 mL of distilled water using a 10 mL graduated cylinder, and pour into the plastic cup containing the pre-weighed KI for the trial. Swirl to dissolve the KI. Repeat twice, so that there are 10 mL of KI solution.
4. Start the data collection and collect a 3 second baseline with the flask sealed. Then, remove the rubber stopper and quickly add the KI solution into the reaction flask, and immediately seal the flask.
5. Swirl the flask for *5 seconds*, then keep the flask still for the remaining time.
6. After data collection ends, save the run, and record the LabQuest run number with the corresponding trial.
7. Dispose the solution into the waste beaker, and rinse the reaction flask once using tap water, then rinse again with distilled water.
8. Repeat *steps 1-7* for each trial at each temperature level.

2.2.5 Cleanup.

1. Disassemble the apparatus.
2. Save the file on the LabQuest.
3. Neutralize the waste solution by adding a small amount of sodium thiosulfate, then dispose the waste solution down the drain with running water.

3 Data Collection (67 words)

Level	Trial	Target / °C	Run	Hot plate / °C	Start temp / °C (± 0.5 °C)	End temp / °C (± 0.5 °C)	Mass of KI / g (± 0.01 g)
1	1	15	4	45	14.4	15.1	0.30
	2		5		15.4	16.0	0.29
	3		6		14.7	14.0	0.32
	4		7		14.9	15.3	0.32
	5		8		15.32	14.4	0.31
2	1	20	48	55	19.3	18.7	0.30
	2		9		21.4	20.5	0.28
	3		10		21.0	21.3	0.29
	4		11		19.7	19.4	0.31
	5		12		19.3	19.8	0.31
3	1	25	13	65	25.2	25.8	0.29
	2		14		25.3	25.8	0.29
	3		15		25.7	26.7	0.29
	4		16		24.7	24.6	0.28
	5		17		25.4	25.9	0.30
4	1	30	18	85	29.6	30.6	0.33
	2		19		29.1	28.5	0.30
	3		20		30.9	31.5	0.30
	4		21		30.0	30.4	0.30
	5		22		30.8	31.0	0.29
5	1	35	23	100	35.6	35.4	0.31
	2		24		35.3	35.8	0.31
	3		25		35.1	33.3	0.32
	4		26		34.7	35.3	0.31
	5		27		35.5	36.3	0.29
6	1	40	43	125	39.8	40.9	0.28
	2		44		41.3	41.1	0.28
	3		45		40.5	37.3	0.29
	4		46		40.0	41.0	0.28
	5		47		39.1	38.8	0.29
7	1	45	33	160	45.6	44.9	0.29
	2		34		44.5	43.2	0.30
	3		35		44.8	44.8	0.29
	4		36		44.5	44.7	0.29
	5		37		45.1	44.2	0.29
8	1	50	38	200	49.1	49.5	0.31
	2		39		49.8	49.0	0.28
	3		40		49.0	48.1	0.29
	4		41		49.2	49.7	0.29
	5		42		49.7	49.9	0.29

Table 1: Data recorded while conducting the experiment.

3.1 Sources of Uncertainty

To reduce random errors, 5 trials were conducted at each temperature level and averaged. Other variables, such as initial H_2O_2 concentration and volume, mass of KI, apparatus setup, and sealing method were all

kept constant across all runs so that differences in the initial pressure-increase rate can be attributed primary to temperature.

4 Data Processing (581 words)

The raw data file can be found in *section A.1*, as the original data obtained from the LabQuest is 300 rows by 80 columns.

Level	Trial	Target / K	Start temp / °C (± 0.5 °C)	End temp / °C (± 0.5 °C)	Average temp / °C (± 0.5 °C)	Volume of H ₂ O ₂ / mL (± 0.06 mL)	Mass of KI / g (± 0.01 g)
1	1	288.15	14.4	15.1	14.75	25	0.30
	2		15.4	16	15.70		0.29
	3		14.7	14	14.35		0.32
	4		14.9	15.3	15.10		0.32
	5		15.32	14.4	14.86		0.31
2	1	293.15	19.3	18.7	19.00	25	0.30
	2		21.4	20.5	20.95		0.28
	3		21	21.3	21.15		0.29
	4		19.7	19.4	19.55		0.31
	5		19.3	19.8	19.55		0.31
3	1	298.15	25.2	25.8	25.50	25	0.29
	2		25.3	25.8	25.55		0.29
	3		25.7	26.7	26.20		0.29
	4		24.7	24.6	24.65		0.28
	5		25.4	25.9	25.65		0.30
4	1	303.15	29.6	30.6	30.10	25	0.33
	2		29.1	28.5	28.80		0.30
	3		30.9	31.5	31.20		0.30
	4		30	30.4	30.20		0.30
	5		30.8	31	30.90		0.29
5	1	308.15	35.6	35.4	35.50	25	0.31
	2		35.3	35.8	35.55		0.31
	3		35.1	33.3	34.20		0.32
	4		34.7	35.3	35.00		0.31
	5		35.5	36.3	35.90		0.29
6	1	313.15	39.8	40.9	40.35	25	0.28
	2		41.3	41.1	41.20		0.28
	3		40.5	37.3	38.90		0.29
	4		40	41	40.50		0.28
	5		39.1	38.8	38.95		0.29
7	1	318.15	45.6	44.9	45.25	25	0.29
	2		44.5	43.2	43.85		0.30
	3		44.8	44.8	44.80		0.29
	4		44.5	44.7	44.60		0.29
	5		45.1	44.2	44.65		0.29
8	1	323.15	49.1	49.5	49.30	25	0.31
	2		49.8	49	49.40		0.28
	3		49	48.1	48.55		0.29
	4		49.2	49.7	49.45		0.29
	5		49.7	49.9	49.80		0.29

Table 2: Raw data.

4.1 Uncertainty Propagation

The uncertainty values for temperature and pressure were obtained from official documentations of the equipments, and the smallest scale on the digital balance was used as the uncertainty of mass: ± 0.01 g. A 25 mL volumetric pipette was used to obtain the H₂O₂ for all trials, but the exact volumes delivered were not confirmed with a graduated cylinder. Thus, the pipette's tolerance was used as the volume uncertainty: ± 0.06 mL. Time was

recorded automatically by the LabQuest, so its uncertainty was assumed negligible relative to pressure variation throughout the experiment.

Tool	Uncertainty
Vernier Stainless Steel Temperature Probe	$\pm 0.1\text{ }^{\circ}\text{C}$ at $0\text{ }^{\circ}\text{C}$, $\pm 0.5\text{ }^{\circ}\text{C}$ at $100\text{ }^{\circ}\text{C}$
Vernier Gas Pressure Sensor	$\pm 4\text{ kPa}$
Digital Balance	$\pm 0.01\text{ g}$
Volumetric Pipette (Class A TD)	$\pm 0.06\text{ mL}$

Table 3: Uncertainty by tool.

However, the methodology in which the data were collected likely increased the uncertainties; for example, temperature varied during the 60 s data collection period, and the probe may not have reflected the solution temperature. Moreover, though marked, the manual insertion depth of the stopper could have introduced small variations. Thus, while linear interpolation with $\Delta T(x) = 0.20 + \frac{0.50-0.20}{100} \cdot x$ could have been used as the temperature probe's uncertainty, a conservative $\pm 0.5\text{ }^{\circ}\text{C}$ was used for all temperature measurements.

4.1.1 Uncertainty in oxygen production rate

For each calculated rate, there are random and instrumental uncertainties. Random uncertainty was propagated using the spread of the five trials' rates at each temperature level using standard deviation, and converted to uncertainty in the mean rate using the standard error of the mean ($\frac{\sigma}{\sqrt{5}}$):

$$\Delta \bar{r}_{\text{rand}} = \frac{\sigma}{\sqrt{n}} \quad (8)$$

Where σ is the standard deviation of the trials. Instrument uncertainty was propagated through the rate calculation:

$$\Delta \bar{r}_{\text{rel, inst}} = \sqrt{\left(\frac{\Delta V_{\text{gas}}}{V_{\text{gas}}}\right)^2 + \left(\frac{\Delta \bar{m}}{\bar{m}}\right)^2 + \left(\frac{\Delta T}{T}\right)^2} \quad (9)$$

$$\Delta \bar{r}_{\text{abs, inst}} = \bar{r} \cdot \Delta \bar{r}_{\text{rel, inst}} \quad (10)$$

The uncertainty of m was taken from the linear regression of the pressure-time data, and the uncertainty of V_{gas} and T were taken from *table 3*. Combined:

$$\Delta \bar{r} = \sqrt{(\Delta \bar{r}_{\text{rand}})^2 + (\Delta \bar{r}_{\text{abs, inst}})^2} \quad (11)$$

4.2 Processed Data

Temp / °C	Trial	Average temp / K (± 0.5 °C)	Mass of KI / g (± 0.01 g)	Start window / s	End window / s	O ₂ production rate / mol s ⁻¹	Avg. rate / mol s ⁻¹	Rate SD / mol s ⁻¹
15	1	287.90	0.30	37.8	60.0	$1.33 \cdot 10^{-6}$	$(1.80 \pm 0.17) \cdot 10^{-6}$	$3.52 \cdot 10^{-7}$
	2	288.85	0.29	47.4	60.0	$1.85 \cdot 10^{-6}$		
	3	287.50	0.32	30.0	60.0	$2.30 \cdot 10^{-6}$		
	4	288.25	0.32	32.0	60.0	$1.86 \cdot 10^{-6}$		
	5	288.01	0.31	29.2	60.0	$1.66 \cdot 10^{-6}$		
20	1	292.15	0.30	31.8	60.0	$1.60 \cdot 10^{-6}$	$(1.80 \pm 0.42) \cdot 10^{-6}$	$9.44 \cdot 10^{-7}$
	2	294.10	0.28	28.0	60.0	$2.49 \cdot 10^{-6}$		
	3	294.30	0.29	25.6	60.0	$2.17 \cdot 10^{-6}$		
	4	292.70	0.31	30.8	60.0	$2.35 \cdot 10^{-6}$		
	5	292.70	0.31	32.8	60.0	$1.90 \cdot 10^{-6}$		
25	1	298.65	0.29	29.8	60.0	$3.74 \cdot 10^{-6}$	$(3.24 \pm 0.28) \cdot 10^{-6}$	$6.09 \cdot 10^{-7}$
	2	298.70	0.29	26.6	60.0	$4.04 \cdot 10^{-6}$		
	3	299.35	0.29	34.2	60.0	$3.05 \cdot 10^{-6}$		
	4	297.80	0.28	24.0	60.0	$2.52 \cdot 10^{-6}$		
	5	298.80	0.30	33.8	60.0	$3.01 \cdot 10^{-6}$		
30	1	303.25	0.33	10.0	60.0	$1.06 \cdot 10^{-5}$	$(7.21 \pm 1.14) \cdot 10^{-6}$	$2.55 \cdot 10^{-6}$
	2	301.95	0.30	13.4	60.0	$5.13 \cdot 10^{-6}$		
	3	304.35	0.30	7.4	60.0	$9.40 \cdot 10^{-6}$		
	4	303.35	0.30	17.0	60.0	$6.20 \cdot 10^{-6}$		
	5	304.05	0.29	19.4	60.0	$5.07 \cdot 10^{-6}$		
35	1	308.65	0.31	7.8	60.0	$9.99 \cdot 10^{-6}$	$(8.21 \pm 0.89) \cdot 10^{-6}$	$1.98 \cdot 10^{-6}$
	2	308.70	0.31	16.8	60.0	$7.61 \cdot 10^{-6}$		
	3	307.35	0.32	15.4	60.0	$8.60 \cdot 10^{-6}$		
	4	308.15	0.31	11.6	60.0	$1.02 \cdot 10^{-5}$		
	5	309.05	0.29	20.6	60.0	$5.28 \cdot 10^{-6}$		
40	1	313.50	0.28	11.2	60.0	$9.54 \cdot 10^{-6}$	$(8.49 \pm 1.16) \cdot 10^{-6}$	$2.58 \cdot 10^{-6}$
	2	314.35	0.28	18.6	60.0	$7.96 \cdot 10^{-6}$		
	3	312.05	0.29	16.2	60.0	$6.06 \cdot 10^{-6}$		
	4	313.65	0.28	8.6	60.0	$1.27 \cdot 10^{-5}$		
	5	312.10	0.29	14.6	60.0	$6.89 \cdot 10^{-6}$		
45	1	318.40	0.29	7.0	60.0	$1.10 \cdot 10^{-5}$	$(1.23 \pm 0.26) \cdot 10^{-5}$	$5.79 \cdot 10^{-6}$
	2	317.00	0.30	16.4	60.0	$9.11 \cdot 10^{-6}$		
	3	317.95	0.29	4.6	60.0	$1.72 \cdot 10^{-5}$		
	4	317.75	0.29	8.6	60.0	$1.96 \cdot 10^{-5}$		
	5	317.80	0.29	12.4	60.0	$5.45 \cdot 10^{-6}$		
50	1	322.45	0.31	10.4	60.0	$2.99 \cdot 10^{-5}$	$(1.76 \pm 0.37) \cdot 10^{-5}$	$8.16 \cdot 10^{-6}$
	2	322.55	0.28	11.8	60.0	$1.12 \cdot 10^{-5}$		
	3	321.70	0.29	14.0	60.0	$1.15 \cdot 10^{-5}$		
	4	322.60	0.29	5.4	52.8	$2.30 \cdot 10^{-5}$		
	5	322.95	0.29	15.2	60.0	$1.35 \cdot 10^{-5}$		

Table 4: Processed data.

Note: The underlined trial was removed from calculations because the source data could not be located on the LabQuest. Run 8 from the LabQuest indicated a target temperature of 293.15K.

4.3 Example Calculation

Consider 15 °C, trial 1 as an example (marked red) in *table 4*. The headspace gas volume was calculated as the difference between the flask volume and the solution volume:

$$V_{\text{gas}} = 250 \text{ mL} - (25 \text{ mL} + 10 \text{ mL}) = 215 \text{ mL} \rightarrow 2.15 \cdot 10^{-4} \text{ m}^3 \quad (12)$$

The analysis window for the pressure-time graph was selected as the main region where pressure increased (from the start of continuous pressure increase to the end of the rise). For this trial, the selected interval was 37.8 s to 60 s:

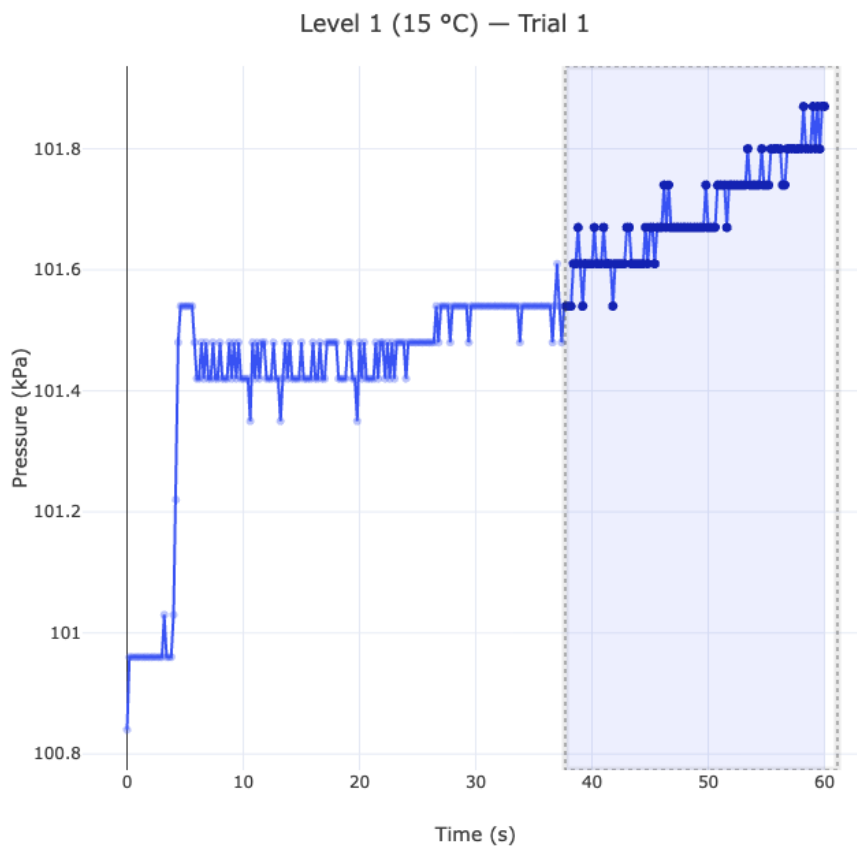


Figure 1: Pressure-time graph.

This window was chosen because the pressure before $t_{\text{start}} = 37.8 \text{ s}$ showed negligible change in pressure, while the pressure after t_{start} increased consistently, due to $\text{O}_{2(g)}$ production. A linear regression with $y = mx + b$ was then fitted to the 111 datapoints in the selected window:

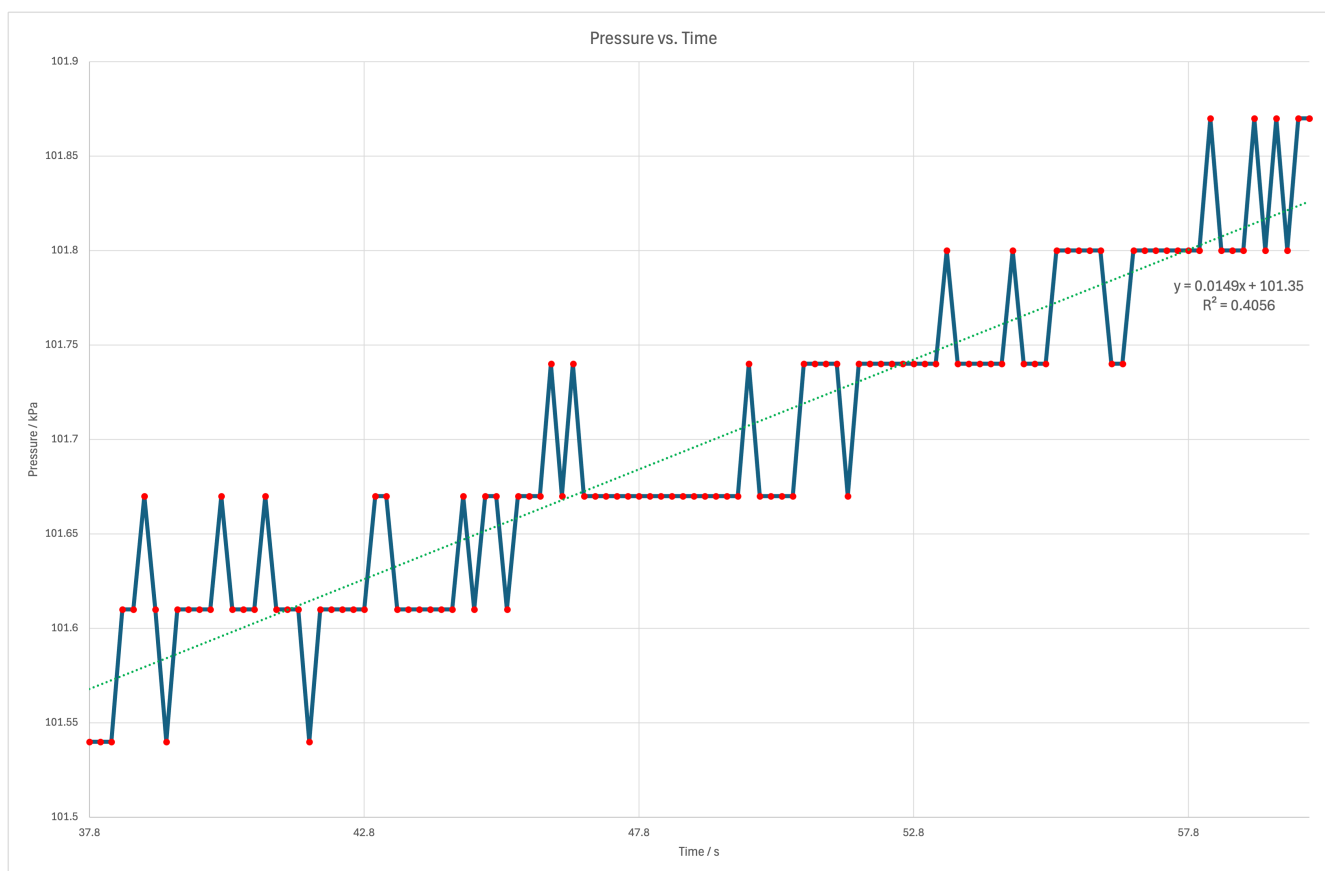


Figure 2: Analysis window.

Note that the slope m was recorded as $0.01486 \text{ kPa s}^{-1}$, which converts to 14.86 Pa s^{-1} . The rate of O_2 production was then calculated from the ideal gas relationship. For a constant headspace volume:

$$r = \frac{\Delta n(\text{O}_2)}{\Delta t} = \frac{V_{\text{gas}}}{RT} \cdot \left(\frac{\Delta P}{\Delta t} \right) = \frac{V_{\text{gas}}}{RT} \cdot m \quad (13)$$

$$r = \frac{2.15 \cdot 10^{-4} \text{ m}^3}{(8.31 \text{ J K mol}^{-1})(288.15 \text{ K})} \cdot 14.86 \text{ Pa s}^{-1} \quad (14)$$

$$r = 1.33 \cdot 10^{-6} \text{ mol s}^{-1}$$

The process above was repeated to calculate the average rate for the other 4 trials at 15°C :

$$\bar{r} = \frac{1}{n} \sum_{i=1}^{n=5} r_i \quad (15)$$

$$\bar{r} = \frac{(1.33 \cdot 10^{-6}) + (1.85 \cdot 10^{-6}) + (2.30 \cdot 10^{-6}) + (1.86 \cdot 10^{-6})}{4} = 1.82 \cdot 10^{-6} \text{ mol s}^{-1} \quad (16)$$

The random uncertainty can then be calculated:

$$\sigma = \sqrt{\frac{1}{4}(-0.49 \cdot 10^{-6})^2 + (0.03 \cdot 10^{-6})^2 + (0.48 \cdot 10^{-6})^2 + (0.04 \cdot 10^{-6})^2} \quad (17)$$

$$\sigma = 3.52 \cdot 10^{-7} \text{ mol s}^{-1}$$

$$\Delta \bar{r}_{\text{rand}} = \frac{\sigma}{\sqrt{4}} = 1.65 \cdot 10^{-7} \text{ mol s}^{-1} \quad (18)$$

Calculating instrumental uncertainty:

1. Average temperature range: $0.4 \text{ }^{\circ}\text{C} \rightarrow \text{half-range} = 0.2 \text{ }^{\circ}\text{C}$

$$\Delta T = \max(0.5, 0.2) = \pm 0.5 \text{ K}$$

$$\therefore \frac{\Delta T}{T} = \frac{0.5 \text{ K}}{288.15 \text{ K}} = 0.001735 \rightarrow 0.174\% \quad (19)$$

2. Average slope uncertainty (from Microsoft Excel's Regression tool): $\pm 0.457 \text{ Pa s}^{-1}$

$$\frac{\Delta m}{m} = \frac{0.457 \text{ Pa}}{14.86 \text{ Pa}} = 0.0307 \rightarrow 3.07\% \quad (20)$$

3. Volume uncertainty: $\pm 0.06 \text{ mL}$:

$$\frac{\Delta V_{\text{gas}}}{V_{\text{gas}}} = \frac{0.06 \text{ mL}}{215 \text{ mL}} = 0.000279 \rightarrow 0.0279\% \quad (21)$$

Combined:

$$\Delta \bar{r}_{\text{rel, inst}} = \sqrt{(0.001735)^2 + (0.0307)^2 + (0.000279)^2} = 0.03075 \rightarrow 3.08\% \quad (22)$$

$$\therefore \Delta \bar{r}_{\text{abs, inst}} = \bar{r} \cdot \Delta \bar{r}_{\text{rel, inst}} = (1.82 \cdot 10^{-6}) \cdot 0.03075 = 5.60 \cdot 10^{-8} \text{ mol s}^{-1} \quad (23)$$

Final uncertainty for the average rate since they are independent uncertainties:

$$\Delta \bar{r} = \sqrt{(1.65 \cdot 10^{-7})^2 + (5.60 \cdot 10^{-8})^2} \quad (24)$$

$$\Delta \bar{r} = 1.74 \cdot 10^{-7} \text{ mol s}^{-1}$$

* * *

The approach from above was used for the other temperature levels:

Temperature level / °C	Temperature level / K	Mean O ₂ production rate, $\bar{r}$ / mol s ⁻¹
15	288.15	$(1.80 \pm 0.17) \cdot 10^{-6}$
20	293.15	$(1.80 \pm 0.42) \cdot 10^{-6}$
25	298.15	$(3.24 \pm 0.28) \cdot 10^{-6}$
30	303.15	$(7.21 \pm 1.14) \cdot 10^{-6}$
35	308.15	$(8.21 \pm 0.89) \cdot 10^{-6}$
40	313.15	$(8.49 \pm 1.16) \cdot 10^{-6}$
45	318.15	$(1.23 \pm 0.26) \cdot 10^{-5}$
50	323.15	$(1.76 \pm 0.37) \cdot 10^{-5}$

Table 5: Mean oxygen production rate.

5 Data Analysis (424 words)

Data from *table 4* and *table 5* were used to create an average initial rate (mol s⁻¹) vs. temperature (K) graph:

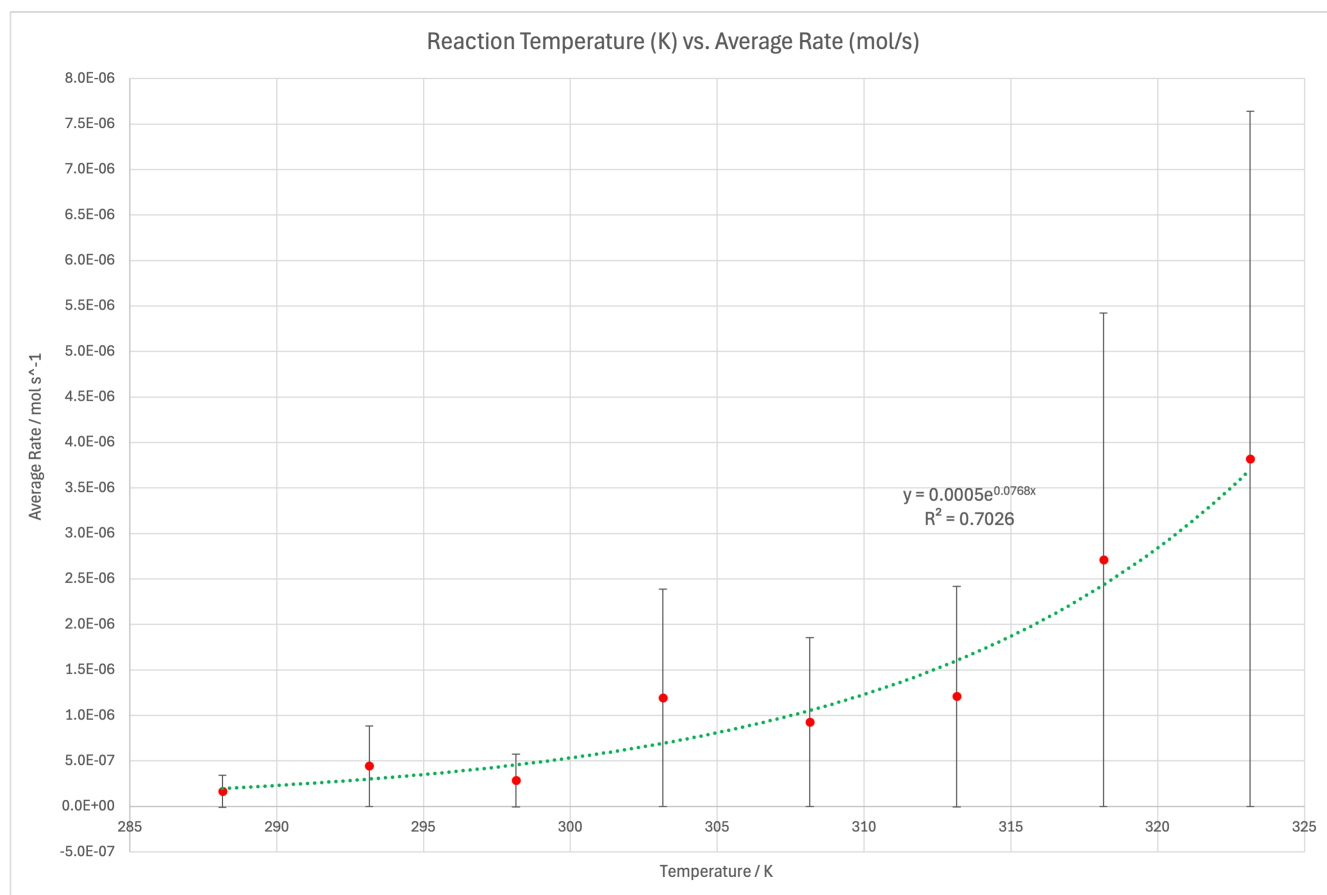


Figure 3: Graph of average rate (mol s⁻¹) vs. temperature level (K).

As shown above, there is an increasing exponential relationship between the rate of O₂ production and the temperature of the system:

$$f(x) = 0.0005e^{0.0768x} \quad (25)$$

This supports the collision theory outlined in *section 1*: higher temperature increases the frequency of collisions and the fraction of molecules with sufficient energy to overcome the E_A ; therefore, the rate of successful reactions are increased.

Given that $[\text{H}_2\text{O}_2]$ and the mass of KI were held constant (minor variations in the mass of KI), the measured rate is proportional to the rate constant by a constant scale factor difference C when initial reactants are constant:

$$r = C \cdot k \quad (26)$$

This allows r to replace k in the Arrhenius equation:

$$\ln(r) = \ln(C \cdot k) = \ln(C) + \ln(k) \quad (27)$$

Thus, plotting $\ln(r)$ against $\frac{1}{T}$ will still yields a straight line with slope $-\frac{E_A}{R}$, only shifting the intercept by $\ln(C)$:

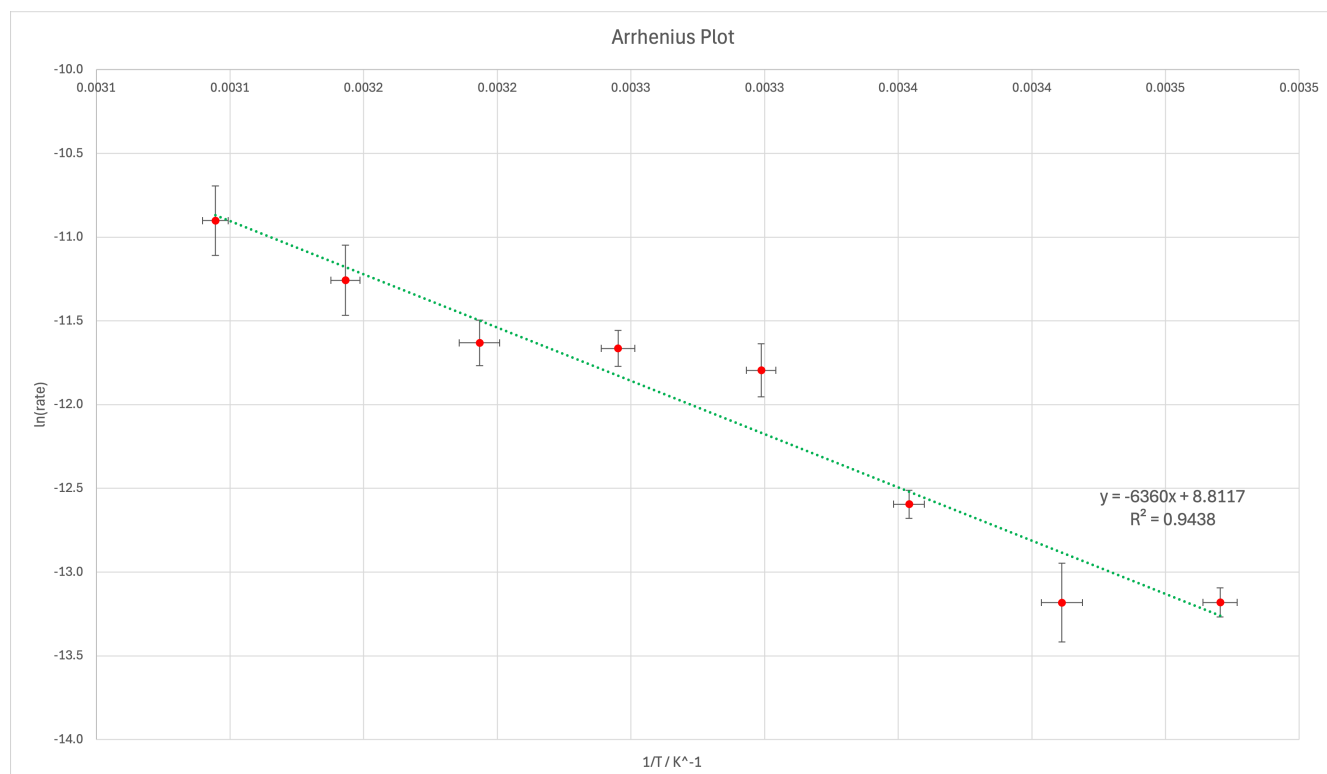


Figure 4: Graph of $\ln(\bar{r})$ vs. $\frac{1}{T}$.

The linear fit above produced a slope of -6360 K and $R^2 = 0.9438$, which indicates a strong linear relationship between $\ln(r)$ and $\frac{1}{T}$.

Recall from *equation 3* that the slope of $\ln(k)$ vs. $\frac{1}{T}$ is $-\frac{E_A}{R}$. Given that $r \propto k$, the slope is used to find the apparent E_A :

$$E_A = |-m \cdot R| = 6360 \text{ K} \cdot 8.31 \text{ J K}^{-1} \text{ mol}^{-1} = 52851.6 \text{ J mol}^{-1} \rightarrow 52.9 \text{ kJ mol}^{-1} \quad (28)$$

$$\therefore E_A = 52.9 \pm 0.6 \text{ kJ mol}^{-1}$$

Although the overall fit is strong, there are small deviations from the trendline that occur at, for example, the fifth datapoint (35 °C). A possible explanation for this is the slightly higher average mass of KI at that level specifically. The average mass was 0.32 g compared to an average of 0.29 or 0.30 g at other temperature levels. A slightly larger KI mass increases the amount of I^- , which can increase the O_2 production rate and $\ln(\bar{r})$ for that level.

In general, the mean O_2 production rate increased from $\bar{r} = (1.89 \pm 0.18) \cdot 10^{-6} \text{ mol s}^{-1}$ at 288.15 K (15 °C) to $\bar{r} = (1.84 \pm 0.38) \cdot 10^{-5} \text{ mol s}^{-1}$ at 323.15 K (50 °C) (*table 5*), which corresponds to an increase by a factor of x :

Calculation:

Uncertainty:

$$x = \frac{\bar{r}_{323.15 \text{ K}}}{\bar{r}_{288.15 \text{ K}}} = \frac{1.84 \cdot 10^{-5}}{1.89 \cdot 10^{-6}}$$

$$\frac{\Delta F}{F} = \frac{\Delta \bar{r}_{323.15 \text{ K}}}{\bar{r}_{323.15 \text{ K}}} + \frac{\Delta \bar{r}_{288.15 \text{ K}}}{\bar{r}_{288.15 \text{ K}}}$$

$$\frac{\Delta F}{F} = \frac{0.38}{1.84} + \frac{0.18}{1.89} = 0.302$$

$$x = 9.74$$

$$\Delta F = 0.302 \cdot 9.74 = 2.94$$

$\Downarrow$

$$\therefore x = 9.74 \pm 2.94 \quad (29)$$

This increase by almost 10 times shows that higher temperature does increase the rate constant. Because of the deviation in the fifth datapoint (explained earlier), a graph of the mass of KI vs. rate per trial was plotted to check whether small variations in the mass of KI had an effect on the reaction rate. However, no clear relationship was observed, which shows that the small variations in the mass of KI was not the primary cause of the observed temperature dependence of $\bar{r}$, and that the higher average KI mass in the 35 °C was only regional to that level.

6 Conclusion (246 words)

This investigation aims to examine the relationship between reaction temperature and the apparent rate constant for the KI-catalyzed decomposition reaction of H_2O_2 , determined from the rate of O_2 pressure increase in a sealed flask. The processed data confirms that increasing temperature increases the apparent rate constant, consistent with the Arrhenius model and collision theory. Across the experimented range, the rate increased from $\bar{r} = (1.89 \pm 0.18) \cdot 10^{-6} \text{ mol s}^{-1}$ at 288.15 K (15 °C) to $\bar{r} = (1.84 \pm 0.38) \cdot 10^{-5} \text{ mol s}^{-1}$ at 323.15 K (50 °C), corresponding

to a 9.74 ± 2.94 increase in rate. The Arrhenius plot yielded an activation energy of $E_A = 52.9 \pm 0.6 \text{ kJ mol}^{-1}$ for the catalyzed pathway. Compared to accepted scientific values, the uncatalyzed decomposition of H_2O_2 has an E_A of approximately 75 kJ mol^{-1} , while the catalyzed pathway lowers it to approximately 56 to 56.5 kJ mol^{-1} . However, the experimental value of $52.9 \pm 0.6 \text{ kJ mol}^{-1}$ falls 3 to 3.5 kJ mol^{-1} below the accepted catalyzed range, which could be attributed to the systematic underestimation of the initial rate at higher temperatures: the reaction occurs at a faster rate at higher temperature levels; thus, some $\text{O}_{2(\text{g})}$ could have escaped before the flask was fully sealed. This reduces the pressure-time slope, which compresses the range of $\ln(\bar{r})$ values, leading to a lower E_A .

Overall, this investigation confirms that increasing temperature does increase the measured rate constant for the KI-catalyzed decomposition of H_2O_2 ; the exponential relationship between temperature and rate, the almost 10x increase in rate over a 35°C range in reaction temperature, and the linear relationship depicted in the Arrhenius plot are all consistent with the collision theory and the Arrhenius model, that KI provides an alternative reaction pathway that requires less energy.

7 Evaluation (675 words)

Given that KI was prepared in plastic cups before the experiment, a key strengths lies in pre-sealing each KI portion with plastic wrap and rubber band immediately after weighing. Potassium iodide is hygroscopic ² [1], which means that its mass would increase without proportionally increasing its moles of I^- if left exposed to air. If the pre-weighed KI portions were left unsealed between preparation and the experiment, the effective catalyst concentration would have varied unpredictably between trials, introducing more random variation into the measured rate.

The first weakness was the transfer of H_2O_2 from its original brown bottle into a clear glass beaker during each day of experiment, following the experimental procedure in *section 2.2.2*. The provided H_2O_2 solution was specifically stored in brown bottles to prevent decomposition triggered by UV radiation and exposure to light, which causes H_2O_2 to break down into H_2O and O_2 (not considered during experimental design). The LED lights in the laboratory emit some UV radiation, and exposing the H_2O_2 to this radiation for 1-2 hours per day for over the course of five days could have gradually reduced the concentration of the stock solution within each session. However, because the order of trials was not systematic (order of data collection jumped between temperature levels, due to revisiting earlier levels across different days), the slight decrease in concentration likely did not have an effect on the results. It would have caused random variation to the initial $[\text{H}_2\text{O}_2]$ across trials, which impacted the standard deviation value within each temperature level, rather than causing a systematic bias in the Arrhenius slope. Moreover, as visible bubbling during the experiment was not observed, its effect on the overall uncertainty is likely weak. To eliminate this in future iterations, the H_2O_2 should be pipetted directly from the brown bottle wrapped with foil, or conducting the experiment in a dark, sealed room.

²**Hygroscopic:** a substance that readily absorbs moisture from the air.

The second weakness was the use of two separate stock H_2O_2 solutions. The first bottle was already half-used when received, meaning that it had been opened and stored for an unknown period before this experiment. Given that H_2O_2 degrade over time even when stored in brown bottles [2], the concentration of the first bottle is likely lower than the second new bottle. Given that the duration of which the first bottle was opened before this investigation and which trial used the first bottle is unknown, this weakness introduces random uncertainty in the initial $[\text{H}_2\text{O}_2]$. Moreover, because the rate constant $r = C \cdot k$ depends on the initial $[\text{H}_2\text{O}_2]$ being constant across trials, the measured rate could be driven by not only temperature, but also the initial $[\text{H}_2\text{O}_2]$. To overcome this weakness in the future, only new H_2O_2 stock solutions should be used.

The third weakness was the reaction temperature inside the flask. The solution inside the flask may not have fully equilibrated with the bath during the fitted interval due to the addition of the catalyst (prepared by dissolving solid KI in room-temperature water), which would increase the spread of data in $\bar{r}$. This can be improved by pre-equilibrating the KI solution in the same bath, and measure the temperature inside the flask using a 3-hole stopper with a sealed port for the temperature probe.

The last weakness was caused by the mismatch of the pressure sensor's instrumental uncertainty and the magnitude of the pressure changes measured. The Vernier Gas Pressure Sensor has an uncertainty of ± 4 kPa [3]. However, some trials had a ΔP below 1 kPa. For example, at 15 °C, the average ΔP over the analysis window was approximately 0.4 kPa, which is an order of magnitude below the sensor's uncertainty. Thus, the pressure readings at lower temperature levels would be indistinguishable compared to noise. However, given that linear regression over multiple datapoints averaged out most random noise, this likely had a minor impact on the results relative to other uncertainty sources. To address this in future iterations, a higher-precision pressure sensor or a differential pressure sensor should be used.

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A Appendix

A.1 Complete Raw Data

The Microsoft Excel file for the complete raw data can be found at GitHub: <https://github.com/AccessRetrieved/chemistryia/blob/d6167ade96b497181fbf9c2e57b69c0467dfe979/Data/Data.xlsx>.

A.2 Source Code

All source code and raw data are open-source, available to view on GitHub: <https://github.com/AccessRetrieved/chemistryia>.